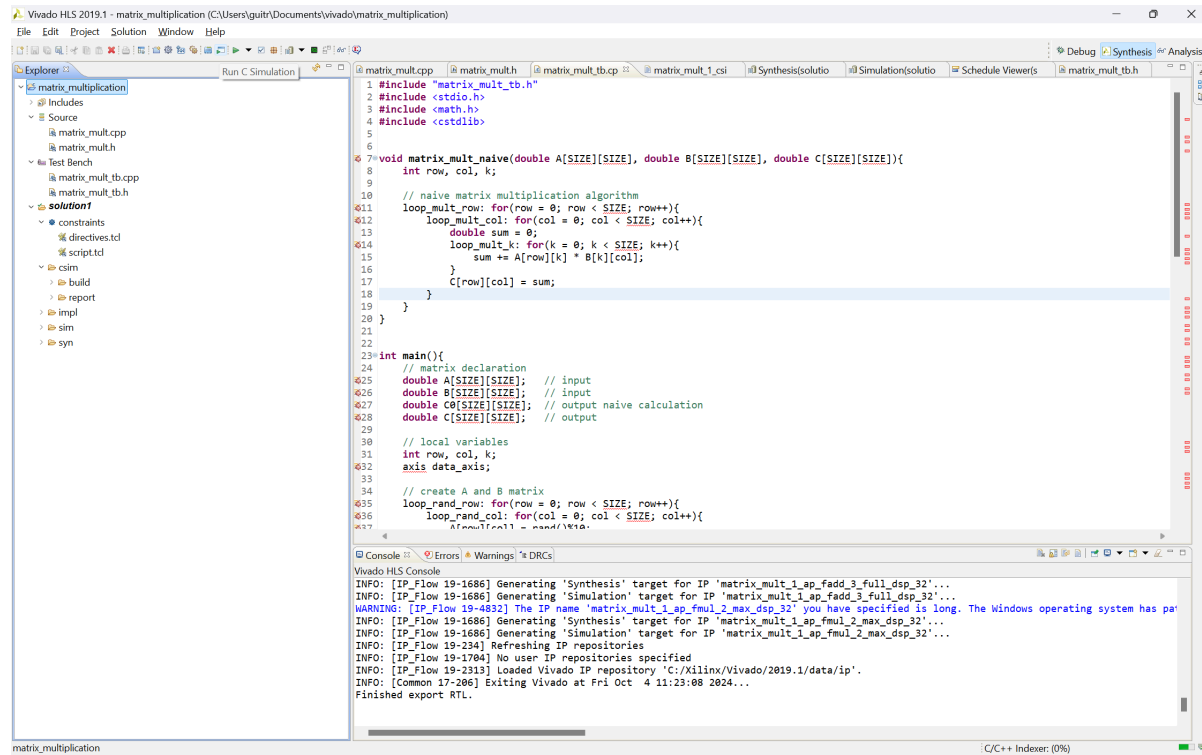


Vivado HLS 2019.1 - Analysis

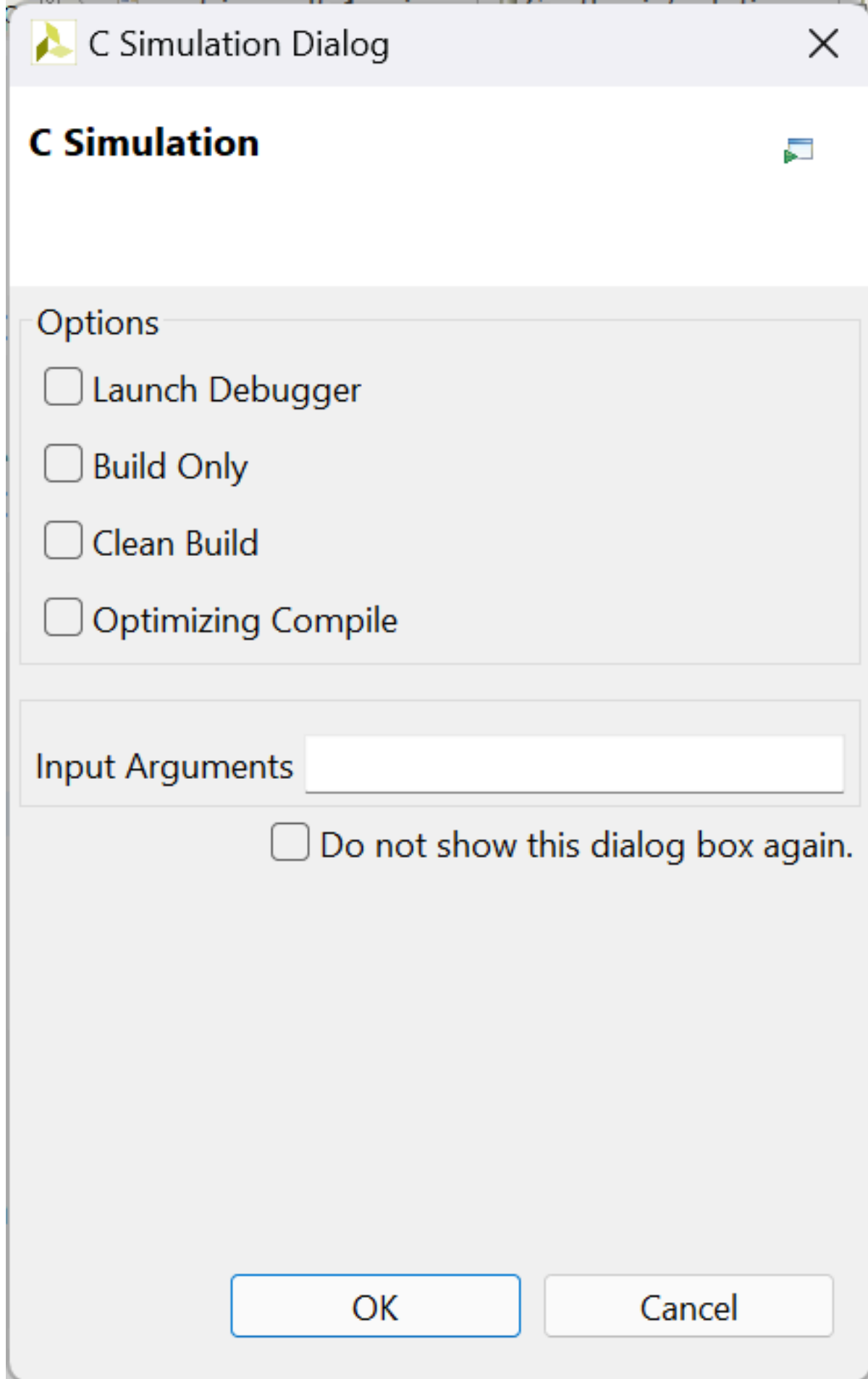
After creating the algorithms it is necessary to evaluate its results.

Run C Simulation

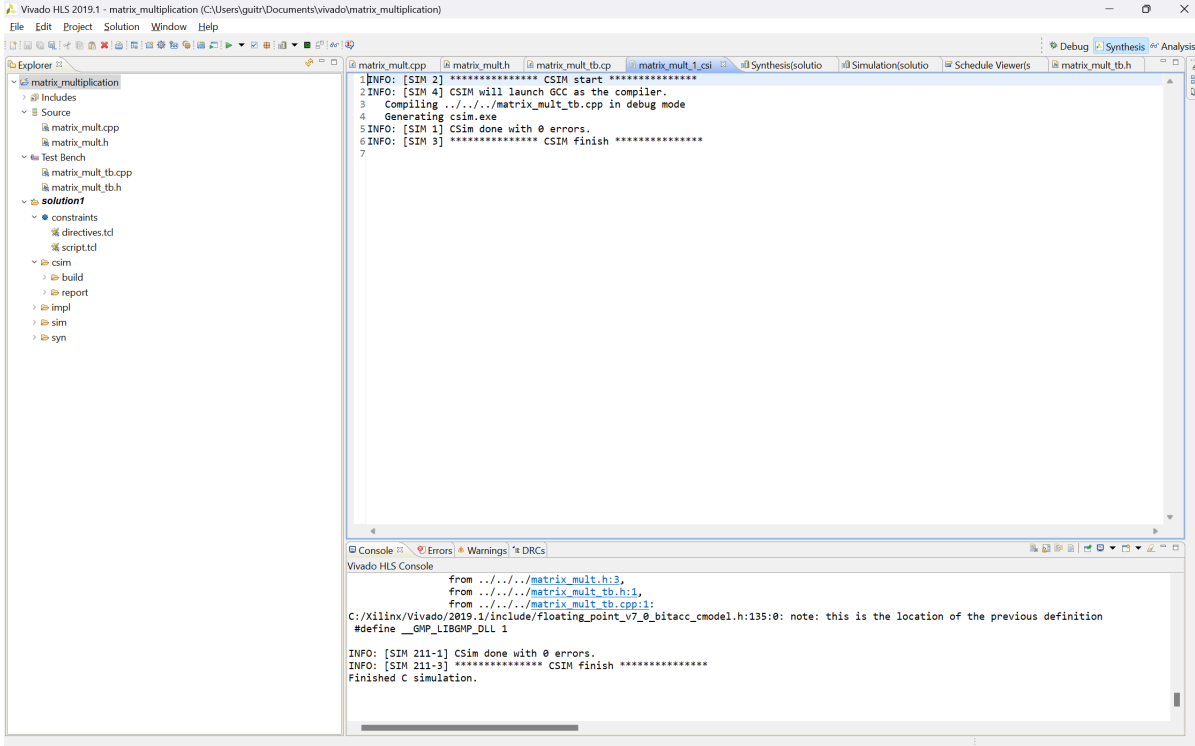
In the upper menu click on the Run C Simulation:



In the options dialog leave it unchanged:



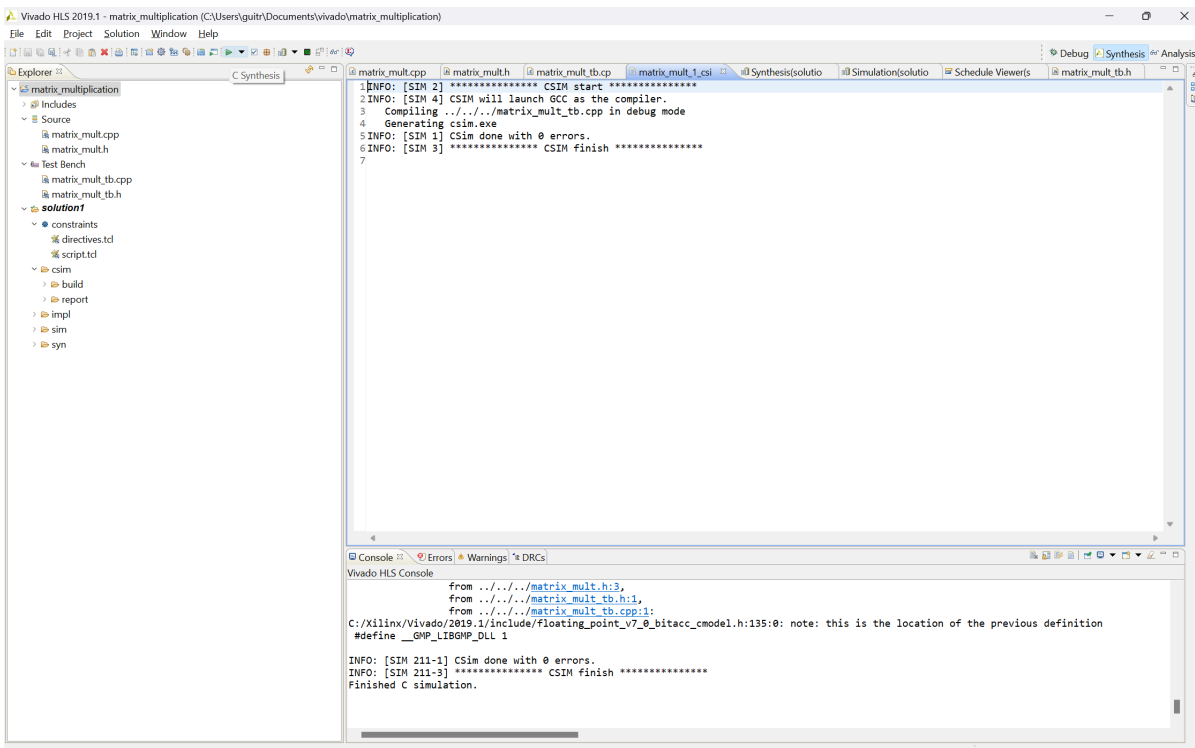
After executing, the following window should appear:



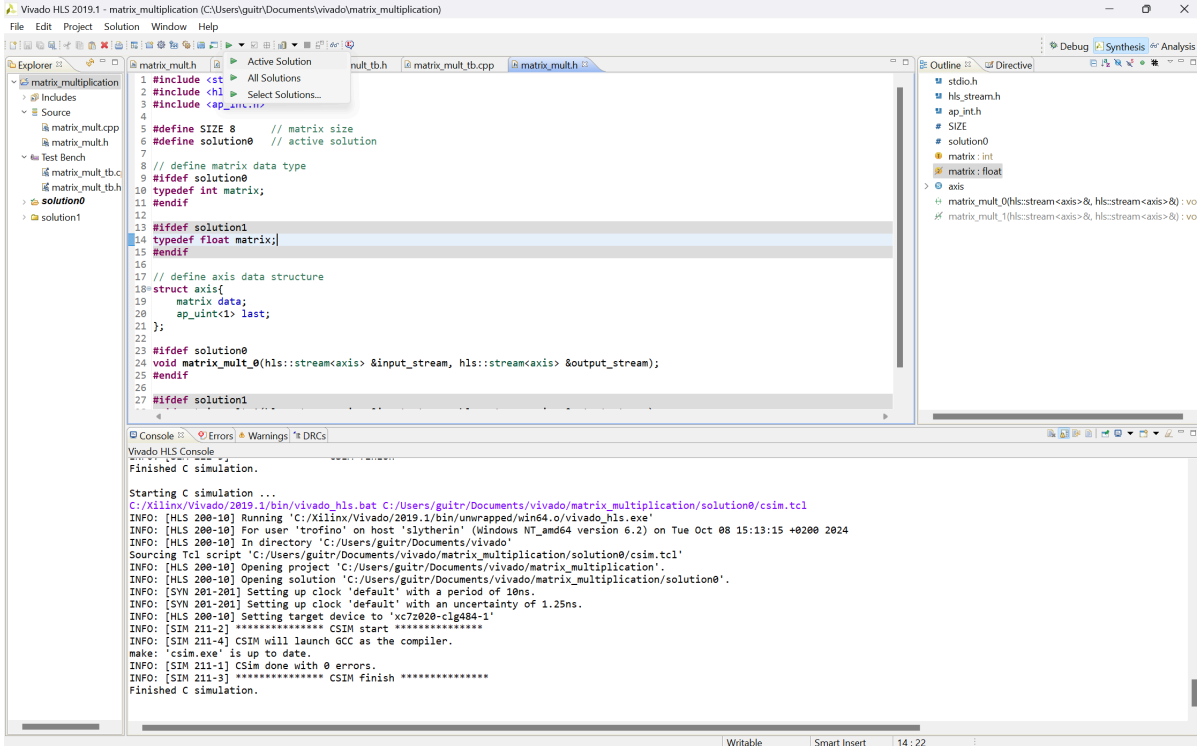
If there are errors, please check [Vivado HLS 2019.1 - Troubleshooting](#).

Run C Synthesis

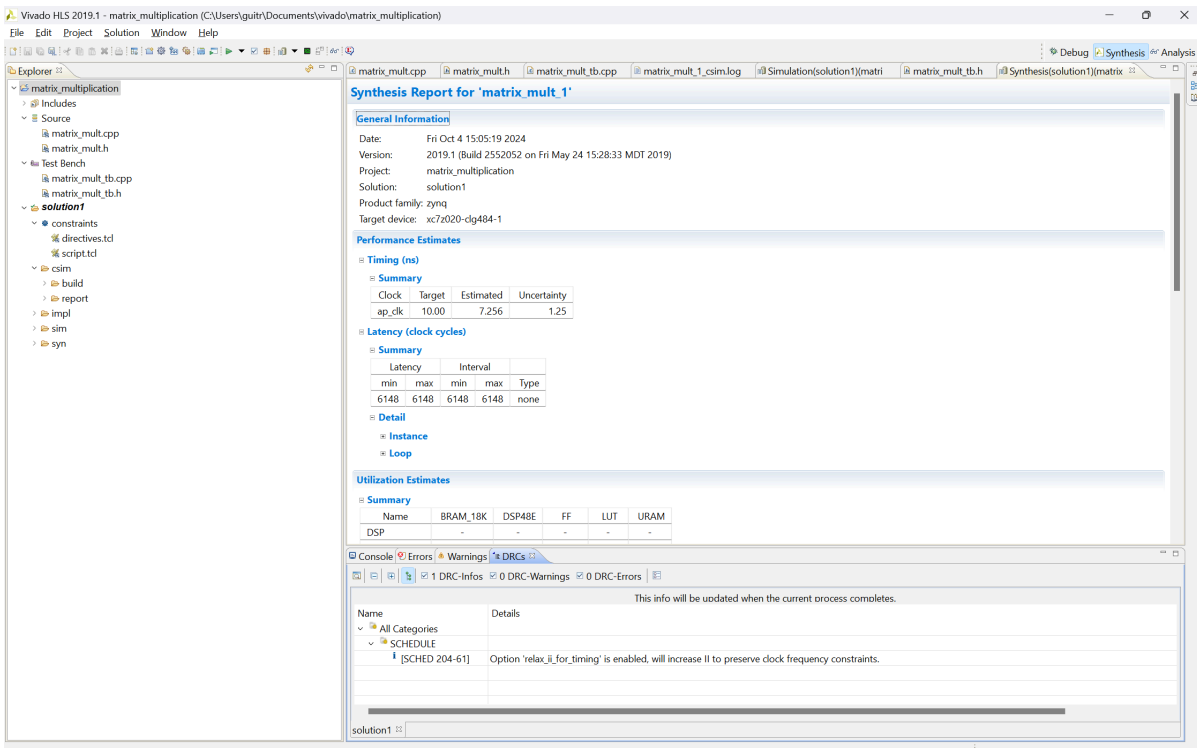
In the upper menu click on the C Synthesis:



Ensure that the correct solution is being used:



After executing, the following window should appear:



Performance Estimates

Timing (ns)

Summary

Clock	Target	Estimated	Uncertainty
ap_clk	10.00	7.256	1.25

Latency (clock cycles)

Summary

Latency		Interval		
min	max	min	max	Type
6148	6148	6148	6148	none

Detail

Instance

Loop

Loop Name	Latency		Iteration Latency	Initiation Interval		Trip Count	Pipelined
	min	max		achieved	target		
- loop_stream_read_row_A	80	80	10	-	-	8	no
+ loop_stream_read_col_A	8	8	1	-	-	8	no
- loop_stream_read_row_B	80	80	10	-	-	8	no
+ loop_stream_read_col_B	8	8	1	-	-	8	no
- loop_mult_row	5776	5776	722	-	-	8	no
+ loop_mult_col	720	720	90	-	-	8	no
++ loop_mult_k	88	88	11	-	-	8	no
- loop_stream_write_row	208	208	26	-	-	8	no
+ loop_stream_write_col	24	24	3	-	-	8	no

Avec l'incertitude par défaut de 1.25 la valeur maximale de clock est de 8.75ns et les opérations avec cette configuration prennent 7.256ns qui est plus petit que la valeur maximale, de cette façon l'opération est possible dans les conditions proposées du système.

La latence représente la quantité de cycles de clock nécessaires pour exécuter les opérations. le tableau d'après montre comment chaque loop contribue avec ce temps. (pour cette raison est important de mettre des flags, maintenant les noms des flags vont apparaître)

Le loop input A2 prend:

- 1 cycle: entrée du loop
- 8 cycles: 1 cycle pour chaque un des 8 interactions (comme la taille de la matrix est 8)
- 1 cycle: sortie du loop

Run C/RTL Co-simulation

Vivado HLS 2019.1 - matrix_multiplication (C:\Users\guib\Documents\vivado\matrix_multiplication)

FileEditProjectSolutionWindowHelp

Run C/RTL Cosimulationmatrix_mult.cppmatrix_mult.hmatrix_mult_tb.cppmatrix_mult_1_csim.logSimulation(solution1)(matrix_mult_tb.hSynthesis(solution1)(matrix_mult_1_csim.log)Analysis

matrix_multiplication

- Includes
- Source
 - matrix_mult.cpp
 - matrix_mult.h
- Test Bench
 - matrix_mult_tb.cpp
 - matrix_mult_tb.h
- solution1
 - constraints
 - directives.td
 - script.td
 - csim
 - build
 - report
 - impl
 - sim
 - syn

Synthesis Report for 'matrix_mult_1'

General Information

Date:

Fri Oct 4 15:05:19 2024

Version:

2019.1 (Build 2552052 on Fri May 24 15:28:33 MDT 2019)

Project:

matrix_multiplication

Solution:

solution1

Product family:

zynq

Target device:

xc7z020-clg484-1

Performance Estimates

Timing (ns)

Summary

Clock	Target	Estimated	Uncertainty
ap_clk	10.00	7.256	1.25

Latency (clock cycles)

Summary

Latency		Interval		Type
min	max	min	max	
6148	6148	6148	6148	none

Detail

Instance

Loop

Utilization Estimates

Summary

Name	BRAM_18K	DSP48E	FF	LUT	URAM
DSP	-	-	-	-	-

ConsoleErrorsWarningsDRCs

This info will be updated when the current process completes.

Name

Details

All Categories

SCHEDULE

[SCHED 204-61]

Option 'relax_ii_for_timing' is enabled, will increase II to preserve clock frequency constraints.

solution1



C/RTL Co-simulation



Verilog/VHDL Simulator Selection

Auto



RTL Selection



Verilog



VHDL

Options



Setup Only

Dump Trace

all



Optimizing Compile



Reduce Diskspace



Wave Debug



Disable Deadlock Detection



Do not show this dialog box again.

OK

Cancel